

AeroVis3R: Geometry-Aware Vision–Language Models for 3D Reasoning over UAV Landmark Videos

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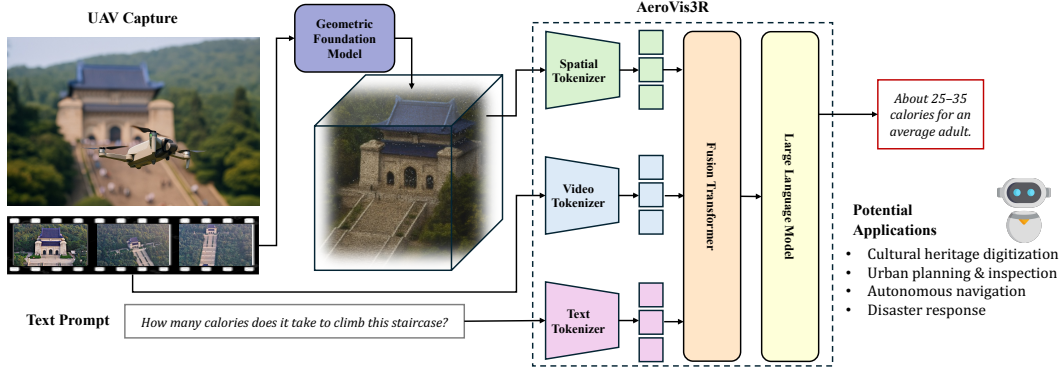


Figure 1: AeroVis3R pipeline. UAV landmark videos are reconstructed into 3D representations and distilled into spatial tokens. Together with video and text tokens, they are fused by a transformer and processed by a pretrained VLM, enabling various applications.

Abstract

Unmanned aerial vehicles (UAVs) enable scalable capture of cultural landmarks, yet monocular video alone remains geometrically ambiguous and difficult for language models to interpret. We present **AeroVis3R**, a unified framework that couples geometric reconstruction with multimodal reasoning. Our system reconstructs dense 3D structures from UAV videos and distills them into compact geometry representations that can be jointly processed with visual and textual inputs. To support research in this direction, we collect a new dataset of 92 landmark videos paired with automatically aligned textual descriptions. By integrating explicit geometry into the reasoning process, **AeroVis3R** achieves geometry-grounded multimodal understanding that bridges UAV video capture, 3D reconstruction, and language-based analysis. The dataset will be made publicly available upon acceptance to encourage further advances in geometry-aware multimodal learning.

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1 Introduction

Unmanned aerial vehicles (UAVs) provide a low-cost way to capture landmark videos at scale, but monocular footage remains geometrically ambiguous. While large multimodal models can now interpret and reason over visual input, they largely operate in 2D and struggle with tasks requiring explicit spatial understanding.

We present **AeroVis3R**, a framework that bridges UAV video capture, geometric reconstruction, and multimodal reasoning. By distilling reconstructed 3D structures into geometry-aware embeddings and combining them with video and text, AeroVis3R enables queries that go beyond surface appearance. Alongside the model, we release a dataset of 92 UAV landmark videos with aligned text annotations, offering a benchmark for geometry-grounded multimodal research with applications in cultural heritage, tourism, and navigation.

2 AeroVis3R Framework

The overall pipeline of **AeroVis3R** (Fig. 1) consists of four stages, from UAV video capture to multimodal reasoning. UAVs record monocular videos of landmarks from diverse viewpoints, providing the raw input sequence $\{I_t\}_{t=1}^T$ for geometric reconstruction. We adopt a geometry foundation model F_{geo} and extract intermediate tokens generated before its estimation heads. These *geometry tokens* $\mathcal{G} = \{g_i\}_{i=1}^{N_g}$ are obtained as $\mathcal{G} = F_{\text{geo}}(\{I_t\}_{t=1}^T)$, where each $g_i \in \mathbb{R}^d$ encodes spatial layout, depth, and scale, while also aggregating cross-view correspondences across frames.



Figure 2: (a) UAV and Remote Controller. The drone used for video capture and its handheld controller. **(b) Sample Scenes.** Representative UAV-captured landmark views under different lighting conditions. **(c) Data Statistics.** Distribution of captures across day, dusk, and night (pie chart) and corresponding clip durations (sorted bar chart).

2.1 Multimodal Tokenization.

Each modality is converted into tokens consumable by the reasoning backbone:

- Spatial tokens. Same as geometry tokens $\mathcal{G} = \{g_i\}_{i=1}^{N_g}$, which provide explicit 3D grounding enriched with multi-view consistency.
- Video tokens. The VLM’s native visual tokens $\mathcal{V} = \{v_j\}_{j=1}^{N_v}$, where $v_j = f_{\text{vis}}(I_t)$ for some encoder f_{vis} , summarizing appearance and temporal dynamics from UAV footage.
- Text tokens. The VLM’s native language tokens $\mathcal{T} = \{t_k\}_{k=1}^{N_t}$, where $t_k = f_{\text{xt}}(Q)$ for query Q , encoding user inputs and serving as the semantic interface to geometry.

2.2 Fusion and Reasoning.

The three token streams $(\mathcal{G}, \mathcal{V}, \mathcal{T})$ are concatenated into $\mathcal{X} = [\mathcal{G}; \mathcal{V}; \mathcal{T}] \in \mathbb{R}^{(N_g+N_v+N_t) \times d}$ and processed by a transformer-based fusion module Φ . Geometry tokens act as anchors to guide cross-modal attention between video and text, yielding a unified representation

$$Z = \Phi(\mathcal{X}) = \Phi([\mathcal{G}; \mathcal{V}; \mathcal{T}]),$$

which integrates geometric grounding, visual context, and semantic intent for downstream reasoning.

3 UAV Landmark Dataset

To support geometry-aware multimodal reasoning, we curate a new dataset of landmark videos captured using a DJI Mini 3 Pro drone. Each video is recorded at a resolution of 1280×720 under diverse conditions (daytime, dusk, night), covering a range of cultural and architectural landmarks. Example clips are provided in the supplementary material.

UAVs are flown around landmarks with varying trajectories to maximize viewpoint diversity while remaining within safe flight envelopes. Each clip lasts between 10 and 60 seconds, providing sufficient motion for geometric reconstruction. For each landmark, we align external knowledge from *Wikipedia*, extracting descriptive sentences that are linked to the captured clips. These annotations

enrich the dataset with semantic context, supporting question answering that combines geometric structure and factual knowledge.

As summarized in Fig. 2, the dataset includes (a) the DJI Mini 3 Pro drone and remote controller used in collection, (b) representative sample scenes under varying illumination, and (c) distributions of capture conditions and clip durations. This provides a balanced testbed for evaluating geometry-aware reasoning. The dataset is designed to benchmark UAV-based 3D reasoning tasks. By combining monocular UAV video with aligned textual context, it allows researchers to evaluate systems that must jointly reconstruct, interpret, and reason about landmarks in 3D.

4 Experimental Details

For geometric reconstruction, we adopt VGGT, a geometry foundation model, to obtain intermediate tokens prior to its estimation heads. These geometry tokens provide 3D structural cues that are integrated with other modalities during training.

For fine-tuning the vision–language model, we employed **Low-Rank Adaptation (LoRA)** with an update matrix rank of 128 and an adaptation scaling parameter of 256. The visual and spatial encoders were kept frozen, while a single lightweight attention block was introduced for modality alignment.

Training was conducted for **one epoch** over approximately **1 hour** using **8 NVIDIA A100 GPUs**. This setup allowed efficient adaptation of the initialized backbone (LLaVA-Video-7B-Qwen2) to the UAV landmark dataset while preserving pre-trained knowledge from large-scale video-language instruction tuning.

5 Applications and Outlook

AeroVis3R serves as a foundation for practical deployments and future research. It enables low-cost reconstruction and analysis of cultural landmarks, supports tourism applications where visitors can pose geometry-grounded questions (e.g., “Is this staircase steep?”), and facilitates autonomous navigation by guiding UAVs or ground robots with explicit 3D reasoning. In education, it provides interactive tools for students to explore architectural geometry through natural-language queries. Looking ahead, we plan to extend AeroVis3R with multi-UAV collaborative capture, temporal

evolution modeling of landmarks, and physics-aware reasoning such as stability assessment and accessibility analysis.